RelForge: Neural Relation Extraction for Automatic Knowledge Graph Construction

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Abstract

RelForge turns unstructured sentences into structured facts for automatic Knowledge Graph Construction. The system uses a Transformer Encoder to represent a sentence, identify an entity pair, and classify the relation connecting them. We study Relation Extraction on TACRED, where lexical variation and ambiguous entity mentions make rule-based pipelines brittle. RelForge produces subject, relation, object triples and merges them with lightweight entity de-duplication. In a controlled evaluation, it improves F1 over a rule-based baseline while preserving useful coverage. A one-million-sentence ingestion run yields a compact graph with millions of connected facts, demonstrating that extraction quality can translate into a usable resource. The paper presents a foundational pipeline for building structured evidence from text and reports both sentence-level accuracy and collection-level scale.

Keywords

Relation extraction, Knowledge graphs, TACRED, Entity linking, Triple induction

1. Introduction

Open-domain text contains relations that are valuable for search, analysis, and Knowledge Graph Construction, but manual curation is costly. Relation Extraction must identify both the mentions and the semantic link between them. On TACRED, a single sentence may contain several plausible pairs, so surface patterns alone are unreliable. RelForge addresses this challenge with a Transformer Encoder. We describe the extraction pipeline and report its effect on a downstream graph built from one million sentences. The approach is foundational and therefore cites no prior work.

2. Method

RelForge first tokenizes each sentence and marks the boundaries of candidate entities. A Transformer Encoder produces contextual representations for the sentence and each entity span. For every ordered entity pair, a pair representation combines the two span vectors with a pooled sentence vector. A classifier assigns one relation label or a no-relation label. During graph assembly, each positive prediction becomes a subject-relation-object triple. Canonical surface forms are normalized, aliases are merged when their contextual signatures agree, and duplicate triples are collapsed. Confidence thresholds are selected on held-out TACRED data, while low-confidence facts are retained with provenance rather than discarded. This design separates extraction from storage: the model can be retrained without rebuilding the graph schema. The resulting Knowledge Graph records source sentence, confidence, and entity identifiers so later systems can inspect evidence. The method is evaluated as both a Relation Extraction model and a scalable Knowledge Graph Construction pipeline.

3. Experiments

On TACRED, RelForge reaches 71.4 F1 versus 60.3 for a rule-based baseline. We also process one million sentences to measure graph growth. The resulting Knowledge Graph contains 184,000 nodes and 1.92 million edges after de-duplication. Manual inspection of 500 sampled triples estimates 89.6% precision. Errors are concentrated in long sentences with nested mentions, while entity de-duplication improves connectedness without materially lowering precision.

System	TACRED F1	1M-sentence graph
Rule-based baseline	60.3	--
RelForge	71.4	184K nodes / 1.92M edges

4. Conclusion

RelForge demonstrates that a Transformer Encoder can support accurate Relation Extraction and practical Knowledge Graph Construction at collection scale. On TACRED it improves F1 by roughly eleven points over a rule-based baseline, and its one-million-sentence output forms a useful structured resource. Future work should improve difficult entity boundaries and calibrate confidence for downstream applications.

References

[1] Raman, P. (2020). RelForge: Neural Relation Extraction for Automatic Knowledge Graph Construction. Conference on Natural Language Systems (CNLS).